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COMPARATIVE STUDY OF ALKALI AND SILANE TREATED SUGARCANE BAGASSE FOR MECHANICAL REINFORCEMENT IN EPOXY RESIN NANO COMPOSITES

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ABSTRACT

The nano composite of sugarcane bagasse and epoxy resin were prepared through the hand layup technique by using a glass mold. The fiber was initially treated with caustic soda and silane solution prior to the composite fabrication. Three different polymer composites (untreated, alkali, silane) at six different weight percentage of SCBP reinforcement (0wt%, 5wt%, 10wt%, 15wt%. 20wt% ,25wt%) were fabricated. Tensile, Flexural, Impact and Hardness tests were carried out. Both alkali and silane treated SCBP nano composite have better tensile strength than the untreated composite. The tensile strength and flexural modulus of all the composites decreases slightly as the fiber loading increases. The tensile modulus, flexural strength, impact, hardness of the fabricated composites was found to be increasing with increase in SCBP loading. With the exception of hardness test, the silane treated SCBP nano composites shows better properties than the alkali treated and the untreated SCBP nano composites for all the parameters.

Keywords: Sugarcane Bagasse Powder, Alkali Treatment, Silane Treatment, Epoxy Resin.

1.0 INTRODUCTION

Agriculture is at the heart of sub-Saharan Africa's economic growth. This contributes significantly up to roughly 50% to the GDP in sub-Saharan Africa. Through agro-industries, the refining of crops into a refined product increases value according to Lesego et al. 2016. A significant agricultural crop that is produced all over the world is sugarcane. Sugarcane produces a significant amount of waste. Global socioeconomic development and growth have found this to be relevant. Asia was the original home of sugarcane. Typical weather that is humid and subhumid is ideal for sugarcane growth. (Makul et al., 2016).

By 2050, due to population growth, protecting and enhancing the means of subsistence for small-scale families and tenant farmers has become problematic (United Nations, 2015). This is because unsustainable practices and soil degradation have put agricultural natural resources under duress. The low yield has been documented in the last ten years despite advancements in technology. The high cost of processed and raw food is a result of this worldwide. Consequently, protecting and improving tenant farmers' and small-scale families' means of subsistence has become problematic. (Carvalho et al., 2017). To meet the world's growing and changing need for sustenance, significant advances in resource efficiency and the reduction of waste generated from sugarcane are therefore necessary, and they may even be able to stop environmental damage. More money needs to be spent on research and agriculture, especially in emerging nations. In order to encourage the adoption of sustainable production, this is necessary. This will support the realignment of implicit and explicit agriculture to fulfill the needs of the growing population, as well as farmers, ecosystems, and communities in adapting to reduce and create resilience against the unsustainable agriculture productivity. (Carvalho et al., 2017).

Bagasse comprises roughly 45–50% water, 2–5% sugar that has been dissolved, and 40–45% fibers. After the juice is extracted, it is a byproduct of the sugarcane business. The components of sugarcane bagasse are cellulose (36.0%), pentosans (26.0%), lignin (20.0%), and ash (2.2%). Molasses and mill mud are further by products (Kemausuor et al., 2018). The main chemical constituents of bagasse are cellulose, hemi cellulose and lignin. (Debnath, 2009). Composite materials formed by natural fiber and polymeric matrices constitute a current area of interest in composite research. (Mileo *et al.,* 2011).

There exists an excellent application in fabricating bagasse-based composite toward a wide array of application in building construction such as Roads, Boards and block as reconstituted wood, pump and flowing tiles to mention a few. (Yadav *et al.,* 2015). Sugarcane bagasse is the residue obtained from extracting the sugarcane juice in industries for sugar or wine. (Danso, 2017).

Alkali treatment can lower polymerization, lignin concentration, and hemicellulose in the fiber. Alkaline treatment can affect the number of reaction sites on fiber surfaces and increase surface roughness, leading to improved mechanical characteristics. (Bam *et al.,* 2019). The use of sugarcane bagasse increased the tensile and flexural characteristics of materials. The highest tensile and flexural strength were observed at 15wt% fiber loading. Composites with 15wt% fiber loading demonstrated improved thermal stability compared to other samples. (Jumaidin *et al*., 2021).

Mahesh et al. 2018 reported that the tensile modulus of fiber-reinforced polyester composites treated with silane improved by 150%, whereas the tensile strength of fiber-reinforced polyester composites treated with acetylated and alkali significantly increased. Furthermore, acetylated, permanganate, and silane-treated fiber-reinforced composites showed higher flexural modulus and flexural strength compared to untreated panels.

The investigation of the effect of chemical treatment on the mechanical properties of SCB fibers was achieved by Mansour *et al.* (2021). It explains that when compared to untreated SCB fibers, the alkali treatment of SCB greatly increases the tensile strength and modulus of all fiber-reinforced composites. When compared to untreated SCB fibers, the impact and hardness parameters of the SCB/LDPE composite are also markedly improved by the alkalization treatment of SCB. When fiber content increased from 10% to 30% of the total weight, chemical resistance decreased. In a NaOH solution, composites have demonstrated maximal absorption.

MATERIALS AND METHOD

Table 1: Materials and their sources

SN	MATERIALS	SOURCES

1	Fresh sugarcane bagasse waste	Sugarcane center in samaru, Zaria
2	Epoxy resin (grade 3554A)	Juneng Nigeria Limited, Enugu state
3	Hardner (grade 3554 B)	Juneng Nigeria Limited, Enugu state
4	Distilled Water	Chemistry department, ABU Zaria
5	Glass mold	Polymer and Textile Engineering, ABU Zaria
6	Weighing balance	Polymer and Textile Engineering, ABU Zaria
7	Stirring rod	Polymer and Textile Engineering, ABU Zaria

SAMPLING AND TREATMENT

The sugarcane bagasse was collected from Samaru cane center, Zaria, Kaduna State. It was then washed thoroughly with distilled water to remove both access sugar and dirt particles and then dried in an open air for 48 hours. The dried sample was then grinded to a more smaller particle size and then ball milled for 80 hours using a laboratory mill machine from the department of chemical engineering, Ahmadu Bello University.

The bagasse fiber was then divided into three portions. The first portion remain untreated while the other two portions were subjected to Alkali and silane treatment respectively.

ALKALI TREATMENT

The sugarcane bagasse fiber was treated with 5%wt solution of NaOH for 2 hours at room temperature. The fiber was then rinsed thoroughly with distilled water and dried at room temperature for 3 days. (Vijay *et al.,* 2019).

SILANE TREATMENT

The pretreated bagasse fiber was soaked in a concentration of 5%wt of 3 amino propyl trimethoxy silane solution diluted in water and Acetone. (concentration by 50/50 volume). For 2 hours. The fiber was then dried indirect sunlight for three days after removing it from the solution. (Mylsamy *et al.,* 2020).

COMPOSITE PREPARATION

The composite was prepared in the department of Polymer and Textile engineering laboratory, ABU Zaria. A hand layup technique was adopted by using a glass mold having a size of 200cm by 150 cm by 5cm via manual mixing using a stirring rod. The resin used was epoxy resin which was prepared in the ration of 2:1 amount by weight of epoxy resin and the Hardner. The overall weight of the composite is 100 grams. There was a variation in the composition of the epoxy resin and the sugarcane bagasse fiber. The weight of the sugarcane bagasse which was the reinforcement to that of the matrix was given in table 3.3. The measured matrix was thoroughly mixed in a container and stirred at low speed for 15 minutes to achieve uniformity. The mixture was then poured onto a 5gram SCBP. It was then cured for 24 hours at room temperature and then removed from the mold. To achieve faster and easier removal of the composite from the mold, Vaseline was used as a releasing agent which was applied onto the surface of the mold prior to applying the mixture of the matrix onto it. The fabricated composite was then cut into a suitable dimension to facilitate for further analysis based on the ASTM standards

Table 2: Nomenclature of the SCBP\Epoxy resin composites material

COMPOSITE	RATIO	MATRIX (g) 2:1		SCBP(g)
		Epoxy resin	Hardner	
0wt%SCBP\Epoxy resin	100:0	66.60	33.30	0
5wt%SCBP\Epoxy resin	95:5	63.33	31.66	5
10wt%SCBP\Epoxy resin	90:10	60.00	30.00	10
15wt%SCBP\Epoxy resin	85:15	56.66	28.33	15
20wt%SCBP\Epoxy resin	80:20	53.33	26.66	20
25wt%SCBP\Epoxy resin	75:25	50.00	25.00	25

TEM STUDIES

TEM studies was carried out in accordance with ASTM F1877. For study under an electron microscope, the sample should be transparent to allow the passage of electron beams through it. To achieve this semitransparent nature, the sample is sectioned into fine sections using glass or diamond knife attach to a device known as ultramicrotome. The device has a trough that is filled with distilled water. The section cut is collected in this trough and are then moved to a copper grid to be viewed under the microscope. The size of each section should be between 30nm and 60nm to get the best solution. During preparation of the sample for this study, the sample undergoes fixation, dehydration, infiltration, polishing, cutting and staining

FTIR analysis

FTIR analysis was carried out in accordance with ASTM E1252. FTIR allows the measurements of variation in the fiber composition after chemical treatments. The chemical structure of both pretreated and treated sugarcane bagasse will be evaluated by FTIR spectrophotometer. The sample will be prepared by mixing the material with KBr in a proportion of 1:200 for all spectra. 16 scans will be accumulated within a 4cm resolution. (Cerqueira *et al.,* 2011).

Tensile test

The tensile test was carried out in accordance with ASTM D638 using testicle properties tester. A maximum force of 10KN was applied. The samples to be tested has a dimension of 100mm by 15mm by 5mm of length width and thickness respectively. A crosshead speed of 2mm per minute was used. The specimen was held in the grip of the testing machine and tightened firmly to prevent slippage. The resistance and elongation of the specimen were detected ND recorded by load cell until rupture occurs. Breaking point, elongation at break and tensile modulus were determined and recorded from the test.

Flexural Test

The flexural test was carried out using enerpac universal material testing machine in accordance with ASTM D790. With a maximum load of 100KN and cross bead speed of 5mm/minutes. The samples have a dimension of 100mm by 30mm by 5mm of length width and thickness respectively. The samples were positioned horizontally on the samples compartment of the machine and the test was carried out. Flexural strength and modulus were determined and recorded.

Impact test

The impact test was carried out in accordance with ASTM E23 using the chary impact tester. Samples were tested at room temperature by a single swing of the pendulum hammer using Norwood. The samples dimensions were 100mm by 11mm by 5mm of length width and thickness respectively. Each sample was placed on a vice and clamped firmly. The pendulum hammer was raised to the required height and the released to strike the sample at once. The impact energy absorbed by the sample was recorded. The impact strength of the sample in joules by square meter was recorded.

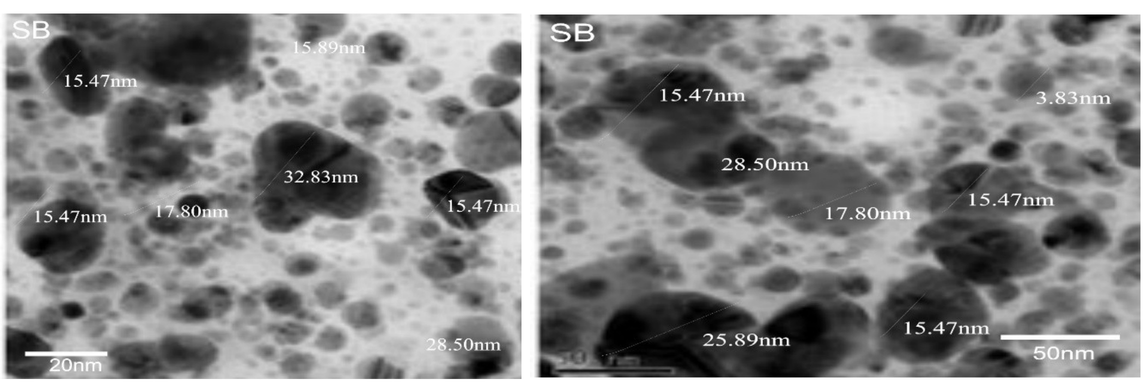
Hardness test

This test was carried out using Vickers hardness tester with maximum capacity of 0.1kgf (150hz) in accordance with ASTM D2220. The test was carried out at temperature 23+2 degrees Celsius. The sample was mounted on the specimen compartment and then the indentation point was focused thereafter. Five different points were indented on each sample and the hardness value were recorded each. The average other the five reading was calculated and recorded. The specimen dimensions were 30mm by 10mm by 5mm of length width and thickness respectively.

RESULTS AND DISCUSSION

TRANSMISSION ELECTRONIC MICROSCOPY (TEM)

The transmission electronic microscope studies as shown in figure 1 shows a representative sample of nano particles with a relatively spherical shape. The particles are mostly dispersed with some aggregation observed. The size distribution is narrow with an average particle size of 19.3nm at a range of 20nm, 50nm and 100nm. This TEM result shows that the nano particles have a uniform size distribution with an average size of 19.3nm. The spherical shape and dispersed state of the particles shows a high degree of uniformity and stability. These results confirm the expected size range of the SCBP which is essential for their intended applications.



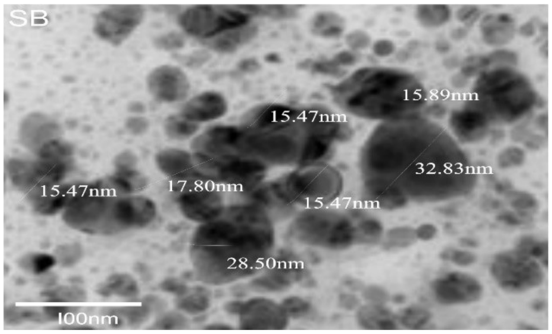


FIGURE 1: TEM image of SCBP at 20nm, 50nm and 100nm

FTIR

Figure 2 shows the FTIR spectra of SCBP of the untreated and the modified alkali and silane treated ones. For the alkali treated SCBP, the spectra show peaks that are attributed to SCBP. For the untreated SCBP, the hemicellulose and lignin show an obvious peak of $1733cm^{-1}$ and $1242cm^{-1}$, the peak for both the hemicellulose and lignin is weaker for the alkali treated SCBP which suggest that they were removed by the Alkali treatment. As for the silane SCBP, its IR spectrum is relatively similar to that of the untreated SCBP bar the little reduction of the hemicellulose and lignin. However, there is an identifiable peak of Si-O-C at 1323.2 cm^{-1}. This bond is formed between the coupling agent and the bagasse. This result is in agreement with Orue *et al.* (2016). They studied, The effect of alkaline and silane treatments on mechanical properties and breakage of sisal fibers and poly lactic acid/sisal fiber composites. Their result show that alkali treatment helps in the removal of hemicellulose and lignin from the sisal fiber and silane treatment modifies the surface properties of the fiber hence improving the strength and stiffness of the fiber. It also improves the compatibility of the fiber with the polymer. Silane treatment did not result in the loss of any peak in the FTIR spectra, the removal of hemicellulose and lignin observed is attributed to the alkali treatment. Si-C peak was not observed in this study most likely due to the low silane concentration used. (Mu et al., 2023).

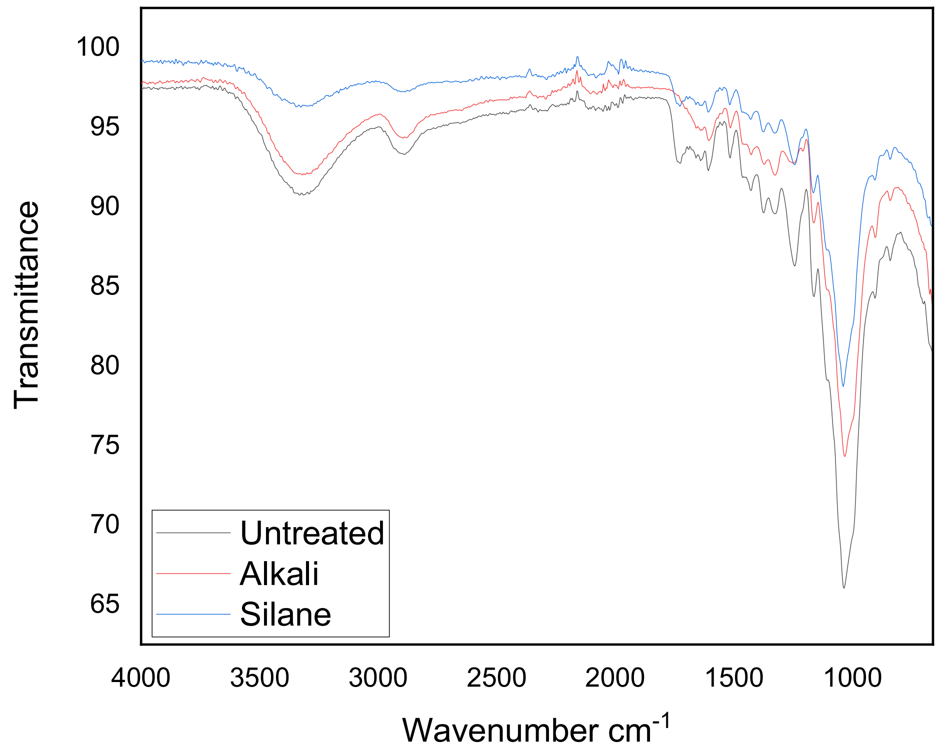


FIGURE 2: FTIR spectra of untreated, alkali and silane treated SCBP.

TENSILE STRENGTH

The tensile strength of Sugarcane bagasse reinforced epoxy nano composite was shown in figure 3 An increase in tensile strength from 5wt%, 10wt%, 15wt% of SCBP was observed for both the Untreated, Alkali and silane treated nano composite. A value of 24.83MPa, 24.99MPa and 30.08MPa were observed for the untreated SCBP, 27.84MPa, 28.06MPa and 31.12MPa for the Alkali treated and 21.92MPa, 25.25MPa and 40.29MPa for the silane treated SCBP nano composite. This is due to reinforcement effect which shows efficient stress transfer from the matrix to the fiber. Beyond the 15wt%, a decrease to 29.63MPa and 26.83MPa for the 20wt% and 25wt% was observed for the untreated, 29.69MPa and 27.1MPa for the Alkali treated and 30.27MPa and 28.26MPa for the silane treated SCBP nano composite. Increase in fiber loading up to 15wt% increases the tensile strength of the composite which shows that the interfacial adhesion between the fiber and the matrix increase simultaneously. It can be seen that there was a sudden decrease in the tensile strength of the SCBP nano composite as the fiber loading increases. This decrease in tensile strength is due to an increase in fiber loading from 15 to 25wt% leading to poor interfacial adhesion between the matrix and the reinforcement (Subramonium *et al.*, 2016). Both alkali and silane treated SCBP nano composite have better mechanical properties than the untreated SCBP nano composite with silane treated SCBP nano composite having the highest tensile strength. This result is in agreement with kanan *et al.* (2023). Who studied the effect of alkali and silane treatments on the polypropylene composite reinforced by marine Posidonia fiber waste. The silane treatment significantly improves tensile characteristics compared to the NaOH treatment. An investigation of tensile strength was carried out, decrease in 5% loading was observed due to poor particles distribution. Later increase in 10% and 15% was observed. Before it decreases again at 20wt% and 25wt%. hence the tensile strength was achieved at 15wt%. (Alewo *et al.,* 2015).

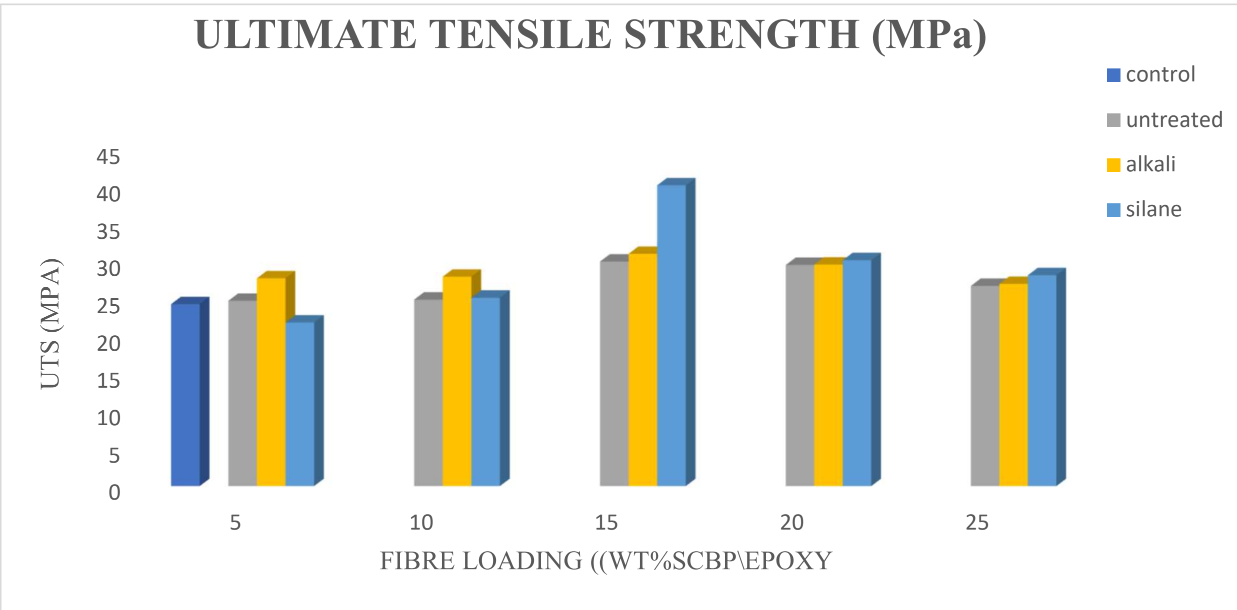


FIGURE 3: TENSILE PROFILE OF SCBP\EPOXY NANO COMPOSITES

TENSILE MODULUS

Figure 4 shows the result for modulus of elasticity. An increase in the modulus of elasticity with increase in fiber load was observed for both the untreated, Alkali treated and silane treated SCBP nano composite respectively. Untreated SCBP nano composite has 2.39MPa for 5wt%, 2.48MPa for 10wt% 2.83MPa for 15wt%, 2.87MPa for 20wt% and 2.93MPa for 25wt%. Alkali and silane treated SCBP nano composite has 2.67MPa and 2.71MPa for 5wt%, 2.79MPa and 2.77MPa for 10wt%, 2.87MPa and 2.92MPa for 15wt%, 3.09MPa and 3.5MPa for 20wt% and 3.19MPa and 3.85MPa for 25wt%. Addition of reinforcement into the matrix increases young modulus which continue to increase with increasing reinforcement. This increase in reinforcement causes the contact surface between the fiber and the matrix to increase, this helps the fiber and the matrix to achieve greater cohesive energy between them while increasing the mechanical properties. It was also found that increase in fiber content increases the tensile modulus of the fiber reinforced composite. (Kaewkuk *et al.,* 2013). Both Alkali and silane treated SCBP nano composite shows improved modulus of elasticity than the untreated SCBP nano composite. Similar trend was observed by Sujaritjun *et al.* (2013). Based on their study on PLA polymer reinforced with Bamboo, grass and coconut fibers, the tensile modulus of the composite increased with increase in fiber loading because incorporation of the fiber improves the stiffness of the material. Increase in tensile modulus of the raffia palm fiber reinforced composite from 0% to 5% concentration was observed and a decrease was observed by increasing the concentration to 10% 15% and 20%. (Anike *et al.,* 2014).

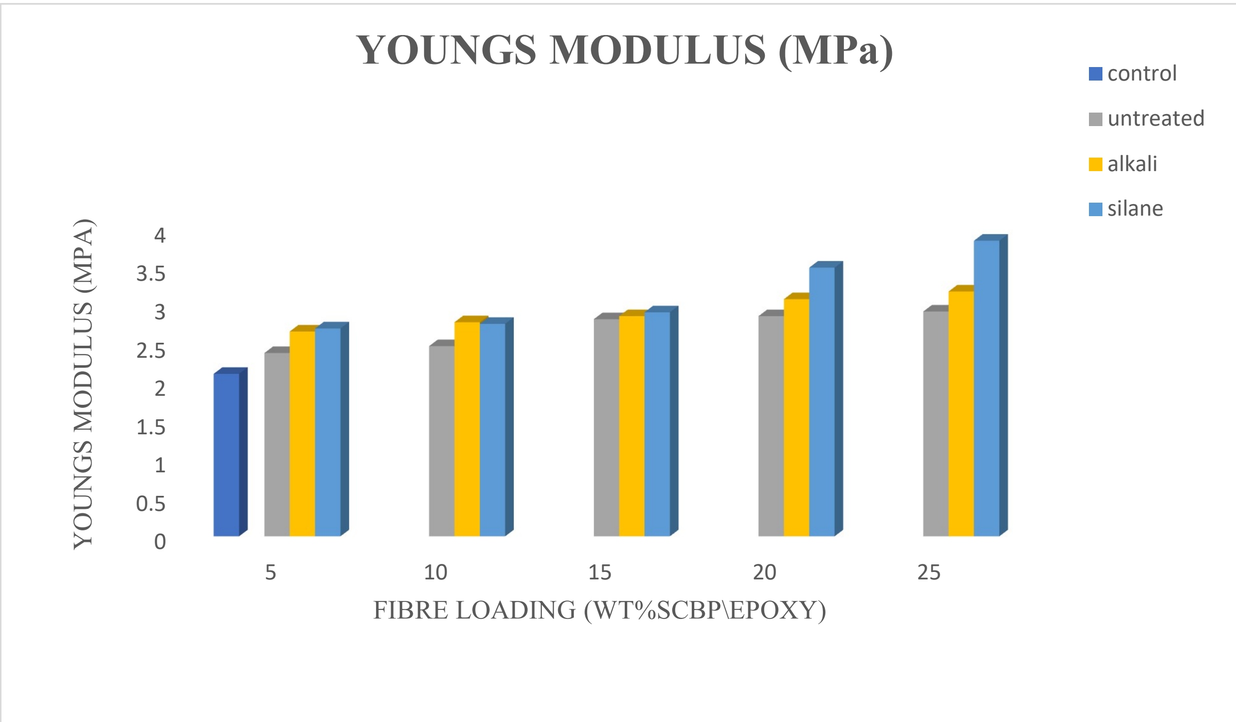


FIGURE 4: YOUNG MODULUS PROFILE OF SCBP\EPOXY NANO COMPOSITES

ELONGATION AT BREAK (%)

Figure 5 shows the elongation properties of SCBP\Epoxy nano composite. Neat SCBP nano composite sample have the highest percent elongation of 12.28%. both the Alkali, silane and the untreated SCBP nano composite have the highest percent elongation of 10.5%, 10.66% and 10.31% respectively at 5% fiber loading and the lowest percent elongation of 9.23%, 9.52% and 9.63% at 25% fiber loading. Hence it can be deduced the elongation properties of the SCBP nano composite decreases with increase in fiber load. Both the Alkali and silane treated SCBP nano composite have higher elongation than the untreated SCBP nano composite. The silane treated SCBP nano composite have the highest elongation than both the alkali treated and the untreated SCBP nano composite. This result follows the same trend with wang *et al.* (2006). on the study of Bamboo fiber composite, a decrease in the elongation of the fabricated composite was observed as the fiber loading increases. This is due toincrease in discontinuity of the polymer matrix as the content of the fiber increases. Surface treatment with benzoyl chloride in PLA/SCBP composites was studied by Yelwa *et al.* (2023). It was observed that there is decrease in the elongation of the composite as the fiber loading increases. The result for elongation at break of bio composite shows that the elongation at break decreases with increase in fiber loading for both the alkali and silane treated composites. (Shehu *et al.,* 2016).

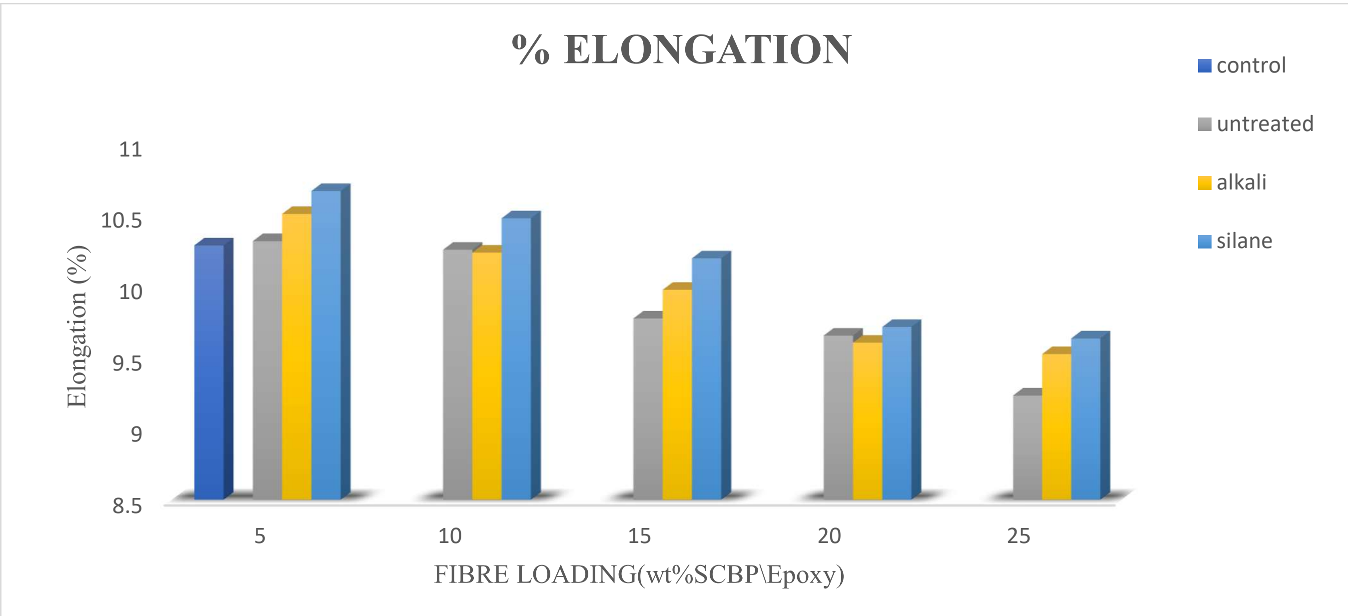


FIGURE 5: ELONGATION (%) PROFILE OF SCBP\EPOXY NANO COMPOSITES.

FLEXURAL STRENGTH

The flexural property of the composite depends on the ductility of the matrix and the strength of the fiber. (Parmiggiani *et al.,* 2021). The flexural strength of the SCBP\Epoxy nano composite was shown in figure 6 Flexural strength of 5wt%, 10wt% and 15wt% were 25.15MPa, 25.76MPa and 31.74MPa found to be increasing with increase in fiber load for the untreated SCBP nano composites. beyond which a decrease in the flexural strength was observed for 20wt% and 25wt% at 29.77MPa and 27.31MPa respectively. Alkali and silane treated SCBP\Epoxy nano composite both shows increase in flexural strength for 5wt%, 10wt% and 15wt% fiber loading at 27.61MPa, 27.87MPa, and 38.36MPa for the Alkali SCBP nano composite treated and 25.4MPa, 26.86MPa and 44.49MPa for the silane treated SCBP nano composite respectively. A drop in the flexural properties was observed for the 20wt% and 25wt% at 35.65MPa and 28.7MPa for the Alkali treated and 37.48MPa and 34.0MPa for the silane treated SCBP nano composite. Both silane and Alkali treated SCBP nano composite have higher flexural strength than the untreated SCBP nano composite which indicates that both the chemical treatments improve the flexural property of the SCBP nano composite. The increase in flexural strength of alkali treated fiber reinforced composite can be due to the fact that the fiber become tougher and more fibrillated than the untreated fiber resulting in a better fiber matrix interface. Sahai et *al.* (2022). in their study of mechanical properties of sisal fiber reinforced polyester composite also found that flexural properties of the sisal composite are enhanced after alkali and silane treatments with silane having the highest flexural strength than the Alkali and the untreated composite.

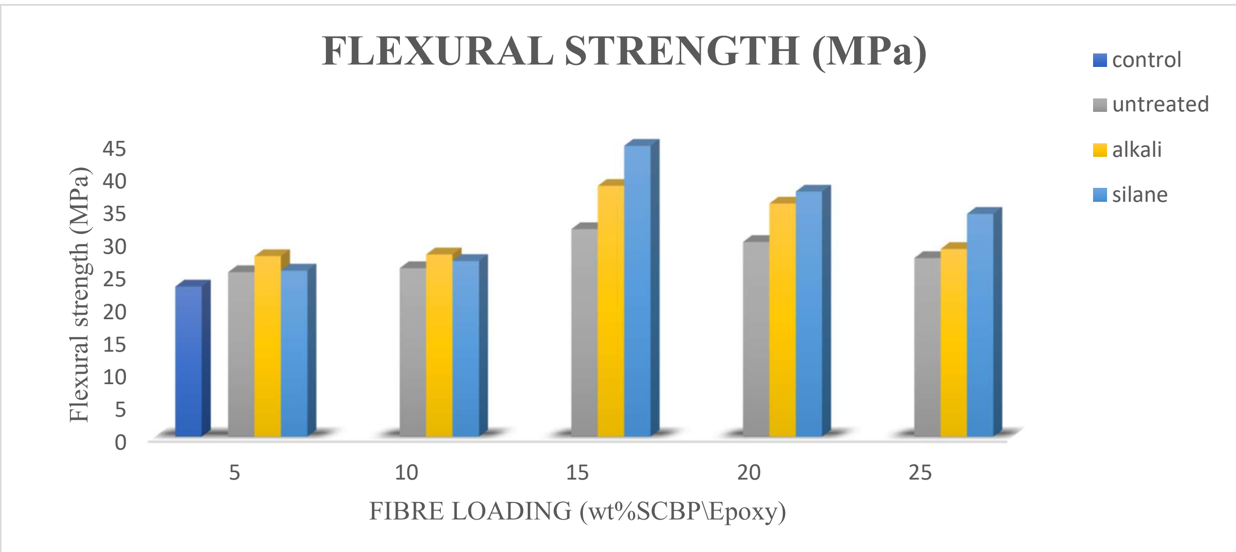


FIGURE 6: FLEXURAL PROFILE OF SCBP\EPOXY NANO COMPOSITES.

FLEXURAL MODULUS

Figure 7 depicts the flexural modulus of untreated, Alkali and silane treated SCBP\Epoxy nano composite. It was found to be 1095.25MPa, 1142.66MPa, 1741.86MPa, 1547.76MPa and 1291.34MPa for untreated SCBP nano composite at 5wt%, 10wt%, 15wt%, 20wt% and 25wt% respectively. It can be observed that the flexural modulus is increasing with increase in fiber loading up to 15wt% before it decreases at 20% and 25% fiber loading. The same trend was found in both the Alkali and silane treated SCBP nano composite with Alkali having 1331.77MPa, 1601.39MPa, 1726.78MPa, 1650.02MPa, 1331.64MPa and silane having 1331.72MPa, 1689.81MPa, 1953.68MPa, 1707.4MPa, 1657.35MPa both for the 5wt%, 10wt%, 15wt%, 20wt% and 25wt% respectively. Alkali and silane treated SCBP nano composite have higher flexural modulus than the untreated SCBP nano composite. Coir fibers in a poly (butylene succinate) matrix for reinforcement. The findings demonstrated that, in comparison to untreated coir fiber-reinforced composites, alkali-treated fiber composites offered greater flexural modulus at all fiber loading levels. The best flexural characteristics were shown by treated fiber composites at 25% fiber loading. (Sever *et al.,* 2011).

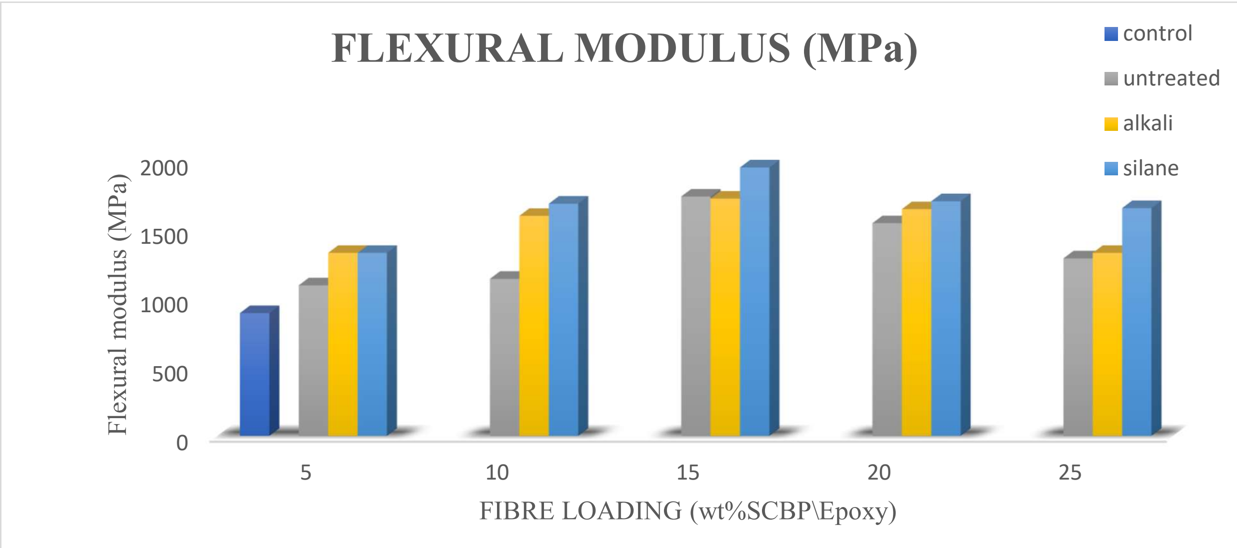


FIGURE 7: FLEXURAL MODULUS PROFILE OF SCBP\EPOXY NANO COMPOSITES.

IMPACT TEST

Impact test was carried out on the SCBP\Epoxy nano composite to study its ability to resist deformation in response to a sudden load.as shown in figure 8, for the alkali treated SCBP nano composite, 0.225 J/m^2 at 5wt% loading, 0.233 J/m^2 at 10wt%, 0.25 J/m^2 and 0.3 J/m^2 at 15wt% and 20wt% and 0.375 J/m^2 at 25wt%. The silane treated SCBP nano composite have 0.2 J/m^2, 0.216 J/m^2, 0.266 J/m^2, 0.3 J/m^2, and 0.333 J/m^2 or the 5wt%, 10wt%, 15wt%, 20wt% and 25wt%. the untreated composite has 0.2 J/m^2, 0.233 J/m^2, 0.25 J/m^2, 0.266 J/m^2, and 0.316 J/m^2 for the 5wt%, 10wt%, 15wt%, 20wt% and 25wt% respectively. It can be observed that all the Untreated, Alkali and silane treated SCBP nano composite increase with increase in fiber loading. Both the alkali and the silane treated SCBP nano composite significantly have higher impact strength than the untreated SCBP nano composite. This shows that both the alkali and silane treatments improve the impact property of the SCBP nano composite. This result is in agreement with the findings of Sahai *et al.* (2022). they found that Alkali and silane treatment both improves the impact strength of sisal fiber reinforced polyester composite. The impact parameter of alkali treated SCB\LDPE was studied, results show improvement in the strength of the fabricated composite as compared to the untreated composite. (Mansour *et al.,* 2021). Dennis *et al.,* (2017). reports that the alkali treated composite have higher impact strength than the corresponding untreated composites.

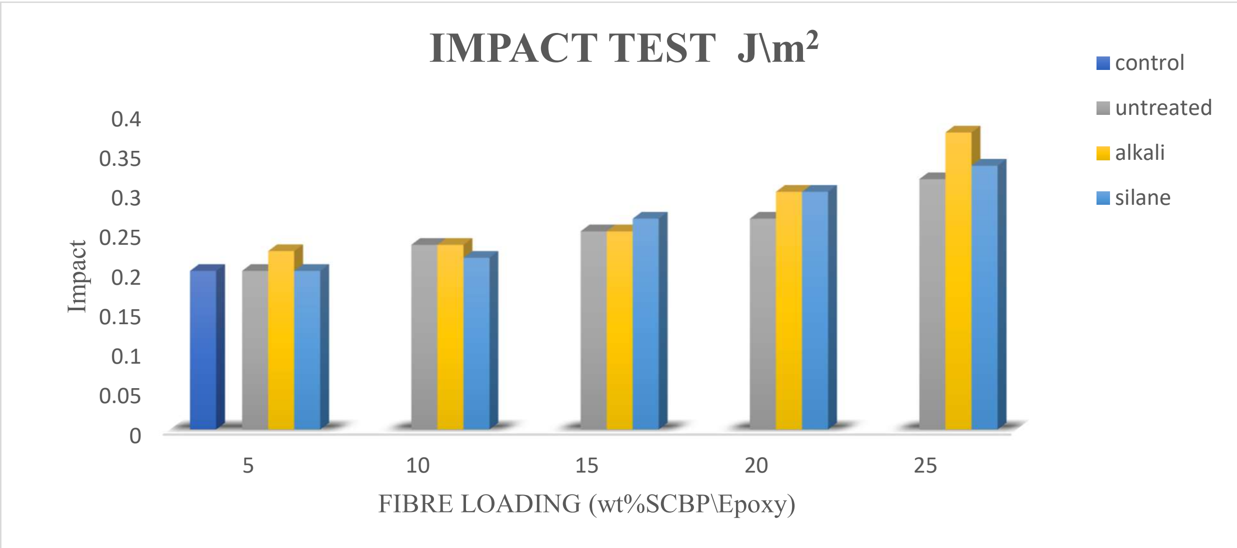


FIGURE 8: IMPACT PROFILE OF SCBP\EPOXY NANO COMPOSITES

HARDNESS TEST

Figure 9 shows the hardness values of SCBP\Epoxy nano composite for the untreated, Alkali and silane treated composites. The untreated SCBP nano composite have 19.5 HV, 24.13 HV, 37.03 HV, 48.73 HV and 57.8 HV for the 5wt%, 10wt%, 15wt%, 20wt% and 25wt%. on the other hand, the alkali treated CBP nano composite have 22.6 HV, 41.7 HV, 47.36 HV, 57.46 HV and 65.86 HV for the 5wt%, 10wt%, 15wt%, 20wt% and 25wt%. it can be observed that both the untreated and alkali treated SCBP nano composite increase with increase in fiber load. Alkali treated SCBP nano composite have better properties than the untreated nano composite. In the case of silane treated nano composite, a similar trend is shown as the hardness keep increasing with increase in fiber loading with the value of 21.4 HV for 5wt%, 28.46 HV for 10wt%, 37.86 HV for 15wt%, 53.16 HV and 54.43 HV for 20wt% and 25wt% respectively. Silane treated nano composite shows better properties than the untreated nano composite at 5%, 10% and 15% fiber loading. But at 25% fiber loading, the untreated appears to have higher fiber loading than the silane nano composite. Alkali treated have higher hardness strength than both the silane and the untreated SCBP nano composite. The mechanical characteristics of polypropylene Bagasse fiber composite was studied by subramonion *et al.* (2016). The result shows that hardness of the fabricated composite was found to be increasing with increase in fiber loading. Chemical treatment on natural fiber disrupt hydrogen bonding withing the cellulose structure which rendered the filler more hydrophobic. This chemical modification promotes mechanical interlocking between fiber and the matrix thus increasing hardness of the composite. From the result in can be deduced that increase in fiber loading for the chemically treated increases the hardness more than the untreated composite. (Anike *et al.,* 2014).

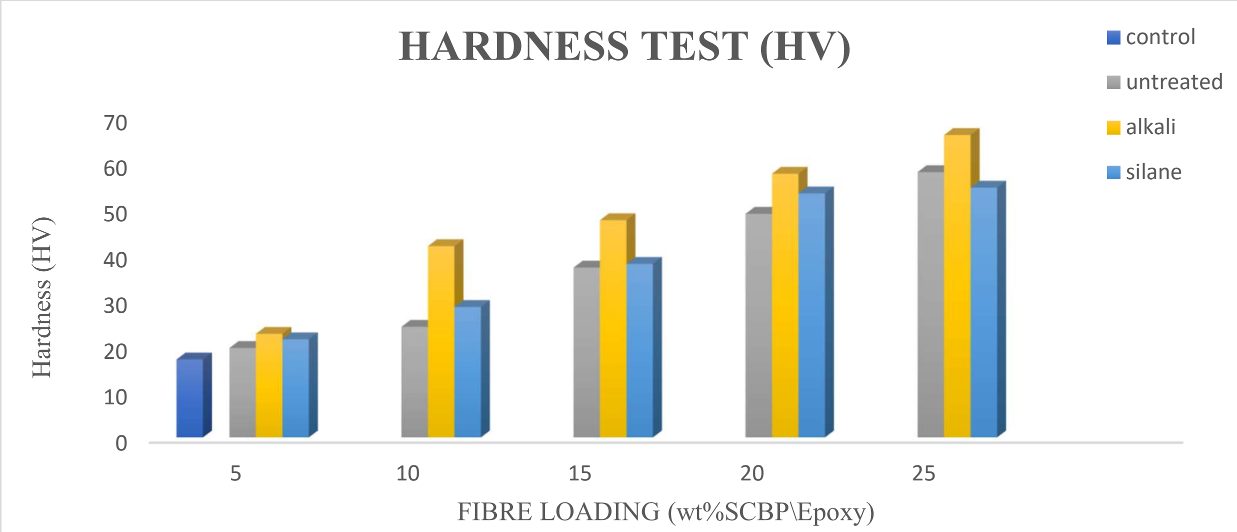


FIGURE 9: HARDNESS PROFILE OF SCBP\EPOXY NANO COMPOSITES

CONCLUSION

The mechanical properties of SCBP\Epoxy resin nano composites for the untreated, Alkali and Silane treatment at different formulations was carried out successfully and the following conclusions were made:

i. The tensile properties of SCBP\Epoxy resin nano composites increase with increase in fiber loading for both the untreated, alkali and silane treated composites up to 15wt%, beyond which it decreases because the matrix have become saturated. Silane treated composites have better tensile strength than the corresponding alkali and untreated composites.

ii. Flexural property of the composites for both the alkali and silane treated showed better bending strength than untreated SCBP\epoxy composite. Silane treated composites have highest impact strength than alkali and the untreated, all of which increases as the fiber loading increase. Hardness follows the same trend as the impact test but alkali treated composite have the highest value than the silane and the untreated.

iii. Despite its significant importance, silane treatment is very costly and that made it less useful as compared with the alkali treatment of the SCBP/Epoxy nano composite.

RECOMMENDATIONS

To further this research work and utilize Sugarcane bagasse powder better, the following recommendations are made:

i. The potential of sugarcane bagasse as reinforcement in another matrix should be investigated.

ii. Other chemical modifications of the sugarcane bagasse powder should be studied on the epoxy resin matrix and another polymer matrix. Examples of chemical treatments include Acetylation, benzoylation, maleated coupling agent to mention a few.

iii. Other characterization such as creep test to show its ability under constant stress, wear test to measure the changes caused by friction, Time temperature superposition, chemical resistance test to study its resistance to Strong acid and bases, thermogravimetric analysis should be carried out to study the longtime performance of the composites.

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